

Date: 2026-06-12

The COSMO® Cube charges exclusively via wireless Qi induction. There is no USB port on the device (by design — for aesthetic and IP54 integrity). The charging pad ships in the box with every unit.

### 3. Transmitter (Desktop Charging Pad)

### 3.1 Technical Specification

| Parameter                | Value                                              |
|--------------------------|----------------------------------------------------|
| Standard                 | Qi 5W (WPC 1.3 certified)                          |
| Form factor              | 80mm × 80mm × 6mm square pad                       |
| Top material             | Matte black TPU (non-slip)                         |
| Bottom                   | Weighted SS base plate (non-slip)                  |
| Transmitter IC           | TI BQ500215 or STMicro STWBC-EP                    |
| Input                    | USB-C, 5V / 2A (10W input max)                     |
| Output efficiency        | ~85% at optimal alignment                          |
| Coil                     | 40mm diameter Qi Tx coil                           |
| LED indicator            | Breathing white (charging), solid green (complete) |
| Foreign object detection | Yes (built-in)                                     |
| Operating temperature    | 0°C to +40°C                                       |
| Cable included           | 1.5m USB-C to USB-A, braided nylon                 |
| Certifications           | CE, FCC (pre-certified OEM module)                 |

### 3.2 LOT Branding on Pad

| Element       | Specification                                |
|---------------|----------------------------------------------|
| Logo position | Top center, 20mm wide                        |
| Logo art      | "LOT®" wordmark in silver metallic ink       |
| Print method  | Silkscreen or pad print                      |
| Bottom label  | "COSMO® Cube Wireless Charger" + safety info |

### 3.3 Sourcing

| Detail            | Value                                                |
|-------------------|------------------------------------------------------|
| Supplier          | OEM Qi pad manufacturer (Alibaba/Shenzhen)           |
| Search terms      | "5W Qi wireless charging pad square OEM custom logo" |
| Quantity          | 100 units                                            |
| Unit price (100+) | ~\$7–\$12                                            |
| Lead time         | 3–4 weeks (with custom branding)                     |
| Certification     | Request CE + FCC pre-certified module                |

## 4. Charging Performance

| Condition | Charge Time |
|-----------|-------------|
|-----------|-------------|

|                             |                                |
|-----------------------------|--------------------------------|
| 0% !' 80% (fast phase)      | ~1.5 hours                     |
| 80% !' 100% (trickle phase) | ~1.0 hour                      |
| 0% !' 100% (full)           | ~2.5 hours                     |
| Maintenance (pad always on) | Safe — BQ25892 manages trickle |

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## 5. Charging Setup Instructions (for Quick Start Card)

1. Place the LOT® charging pad on your desk.
2. Connect via USB-C (5V / 2A minimum – use included cable or any USB-C charger).
3. Place your COSMO® Cube mirror-side DOWN on the pad, centered.
4. LED breathes white = charging. LED solid green = full.
5. Remove when done, or leave on pad – device manages charging automatically.

Note: The Qi receiver coil sits directly behind the mirror-polished back (Side A). That face must contact the pad. Placed this way, the display (Side B) still faces up – notifications remain visible while charging.

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## 6. Revision Notes

v1.1 (2026-07-20): Corrected §5 charging orientation — the v1.0 instructions described the polished (coil) side facing up, which would place the coil away from the pad and prevent inductive coupling. The coil lives behind Side A (mirror-polished back, per COSMO-DEVICE-SPEC-v1.md §4), so Side A must face down onto the pad; Side B (display) faces up, keeping notifications readable while the device charges. No hardware changes — documentation fix only.

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\*Document v1.1 — COSMO® CIA — LOT Systems, Inc.\*  
\*Inventor: Vadim Marmeladov — 2026-06-12, corrected 2026-07-20\*













